



Detergent Presentation - Group 3

BODY WASH

Staying Clean and Healthy!

INTRODUCTION



- Detergents contain surfactants that remove dirt and oil
- body wash (liquid detergent)
- Requires mild, skin-safe, and biodegradable ingredients
- Uses surfactants for cleansing while protecting the skin



The Objectives

- To formulate and produce a body wash detergent
- To understand the function of different surfactants

 Stable foam

 Effective cleansing

 Pleasant fragrance

 Skin-safe pH (5.5–6.0)

Types of Surfactants Used

SLES – Anionic
Main cleanser, Foam

Betaine – Amphoteric
Mildness, Foam booster

Beryl – Non-ionic
Stabilises fragrance,
biodegradable

LABS – Anionic
Improves cleaning power

We can prevent getting sick by doing things like.
Prevention helps our body stay strong like a superhero!

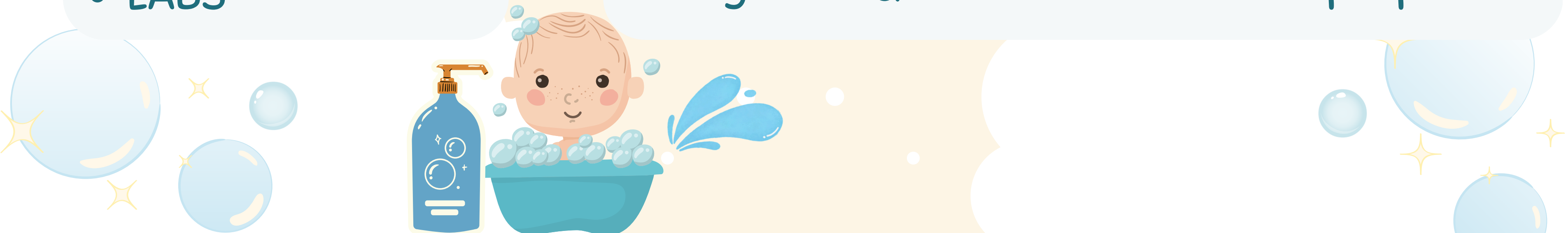
Material Used

Core Surfactants

- Purified water
- SLES
- Betaine
- Berol
- LABS

Supporting Additives (Optional)

- Glycerin – moisturizing agent
- Sodium chloride – viscosity modifier
- Citric acid – pH adjustment
- Preservative – microbial stability
- Fragrance & colorant – aesthetic properties



Formulation

Core Material	Percentage
SLES	12%
Betaine	7%
Berol	2%
LABS	1%
Purified Water	72% (or more, balance formulation to 100%)

Additives	Percentage
Glycerin	3%
Sodium Chloride	0.3 – 1.0%
Citric Acid	0.1 – 0.3%
Preservative	1%
Fragrance & Colorant	0.5%

Example guide : <https://chemistscorner.com/cosmetic-formulations-body-wash/>

Experimental Procedure

1. Water and glycerin were mixed in a beaker.
2. SLES was added slowly with continuous stirring.
3. Betaine and Berol were added to improve foam and mildness.
4. LABS was added in a low concentration.
5. Sodium chloride was added gradually to adjust stickiness.
6. Citric acid was used to adjust pH to 5.5–6.0.
7. Fragrance, preservative, and colorant were added last.

safety Precautions

- Gloves and lab coat were worn at all times.
- Chemicals were handled carefully.
- Avoid direct skin contact.
- pH was controlled to prevent skin irritation.



Results & Discussion

- Clear and homogeneous liquid obtained
 - Stable foam produced
 - Pleasant fragrance achieved
 - Final pH measured between 5.5–6.0
 - Suitable viscosity for body wash
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- Combination of SLES and Betaine provided effective yet mild cleansing.
 - Low concentration of LABS increased cleaning without causing irritation.
 - Berol improved biodegradability and fragrance stability.



Thank You

For Your Attention